

Do We Really Need to Access the Source Data? Source Hypothesis Transfer for Unsupervised Domain Adaptation

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Presenter

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Fundamental Team

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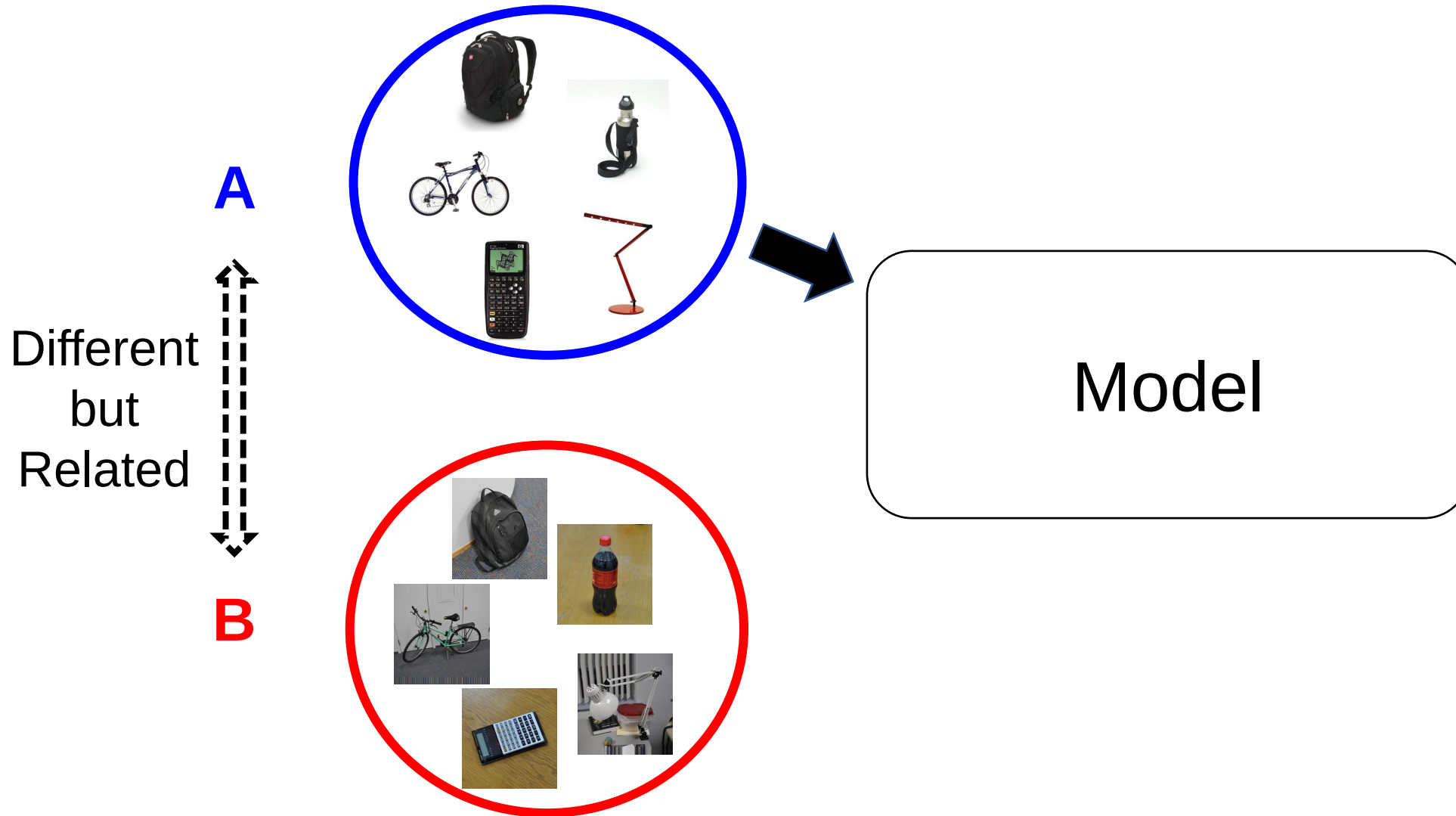
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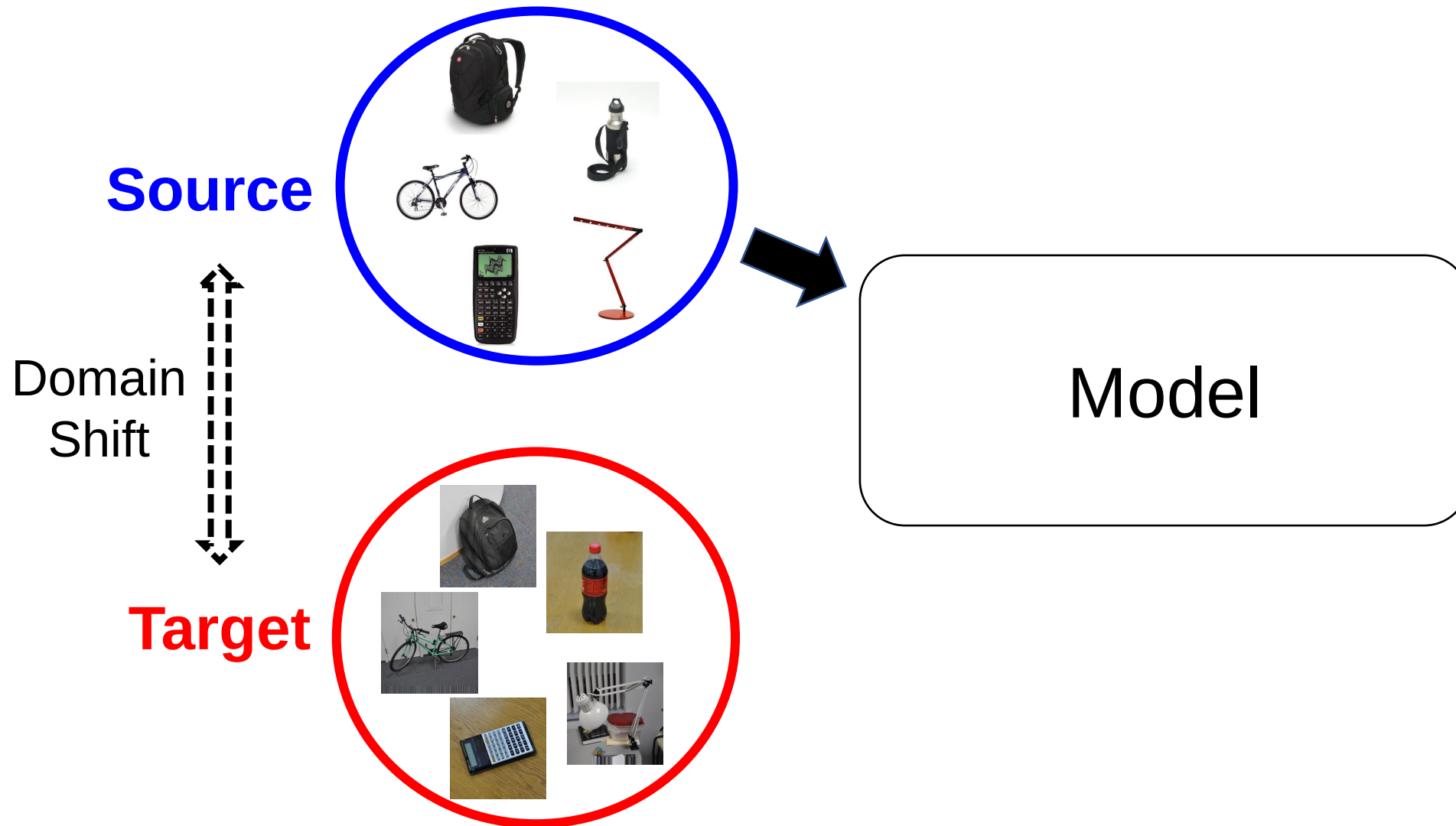
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1. Domain Adaptation

Domain Adaptation



Domain Adaptation



Use Cases

- Apply *AI diagnostic algorithm*, trained on **previous diseases**, to unlabeled data associated with **COVID-19**.
- *A spam filter*, trained on a **certain group of email users** during training, adapt to a **new target user** when deployed.

Domain Adaptation Scenarios

◆ Target Dataset Labeling

- *Supervised* - Labeled Target Data, Much Smaller size compared to Source.
- *Semi-Supervised* - Both Labeled, Unlabeled Target Data.
- *Unsupervised* – Lots of Unlabeled Target Data.

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◆ Domain Relation

- Closed Set – $\{S\} = \{T\}$
- Partial Set – $\{S\} \supset \{T\}$
- Open Set – $\{S\}/\{T\} \neq \emptyset$ and $\{T\}/\{S\} \neq \emptyset$

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Question

2. Background

Domain Adaptation Optimization

Theorem 1

$$\varepsilon_T(h) \leq \underbrace{\varepsilon_S(h)}_{\textcircled{1}} + \underbrace{d_1(D_S, D_T)}_{\textcircled{2}} + \underbrace{\min\{E_{D_S}[f_S(x) - f_T(x)], E_{D_T}[f_S(x) - f_T(x)]\}}_{\textcircled{3}}$$

① : Low error rate in D_S

② : Closer distance between D_S, D_T

③ : label classifier good to both domain

h : hypothesis

ε_T : target error

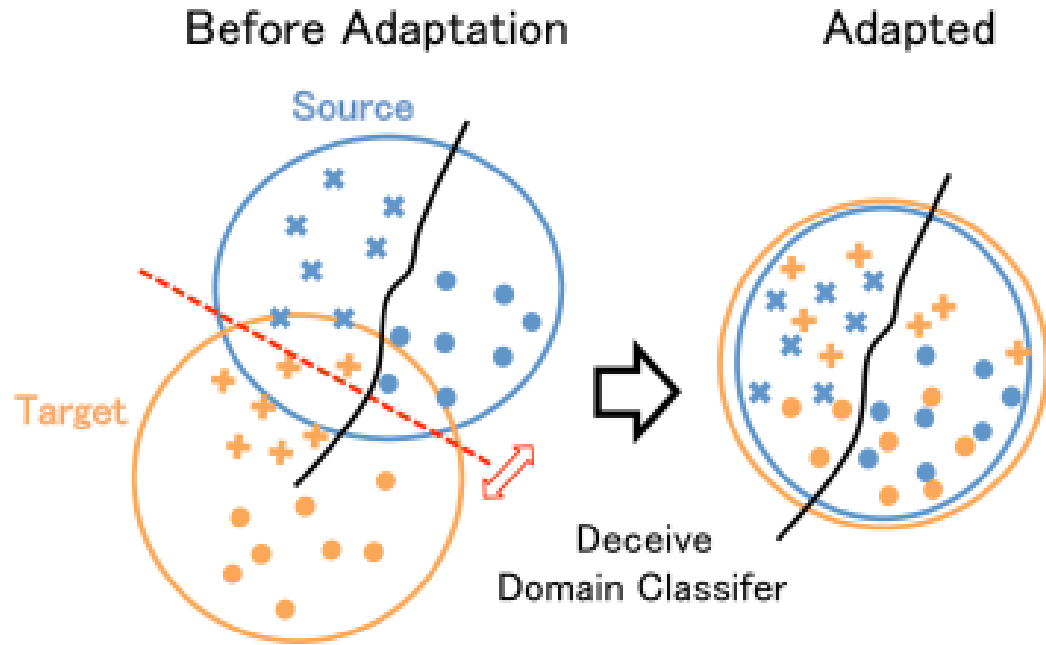
ε_S : source error

$f_S(x)$: source labeling function

$f_T(x)$: target labeling function

$d_1(D_S, D_T)$: L1 – divergence between S, T

Domain Adaptation Task



- ① : Low error rate in D_S
- ② : Closer distance between D_S, D_T
- ③ : label classifier good to both domain

3. Main Idea

Paper Setting

Do we really need to Access the Source Data?

Source Hypothesis Transfer

for Unsupervised Domain Adaptation

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Do we really need to Access the Source Data?

① Inefficient Data Transmission ② Privacy Policy Violation

Source Hypothesis Transfer

for Unsupervised Domain Adaptation

Paper Setting

Do we really need to Access the Source Data?

***S**ource Hypothesis Transfer*

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Paper Setting

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SHOT

- **Theorem 1**

$$\varepsilon_T(\mathbf{h}) \leq \varepsilon_S(\mathbf{h}) + d_1(\mathbf{D}_S, \mathbf{D}_T) + \min\{E_{D_S}[f_S(\mathbf{x}) - f_T(\mathbf{x})], E_{D_T}[f_S(\mathbf{x}) - f_T(\mathbf{x})]\}$$

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⌘ Hypothesis Transfer Learning(HTL)

If D_S and D_T are related, transferred hypothesis does not deviate much from the source hypothesis.

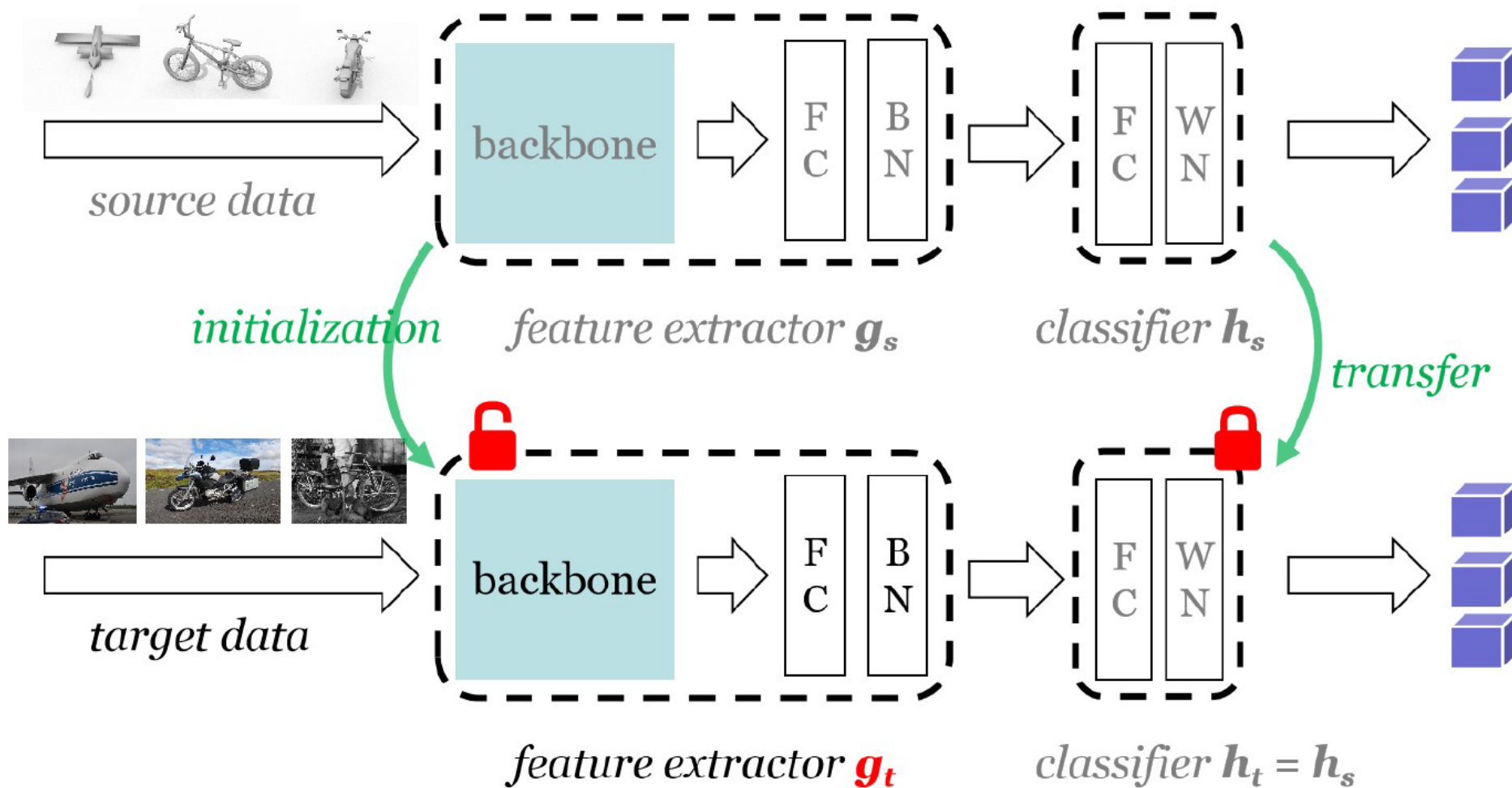
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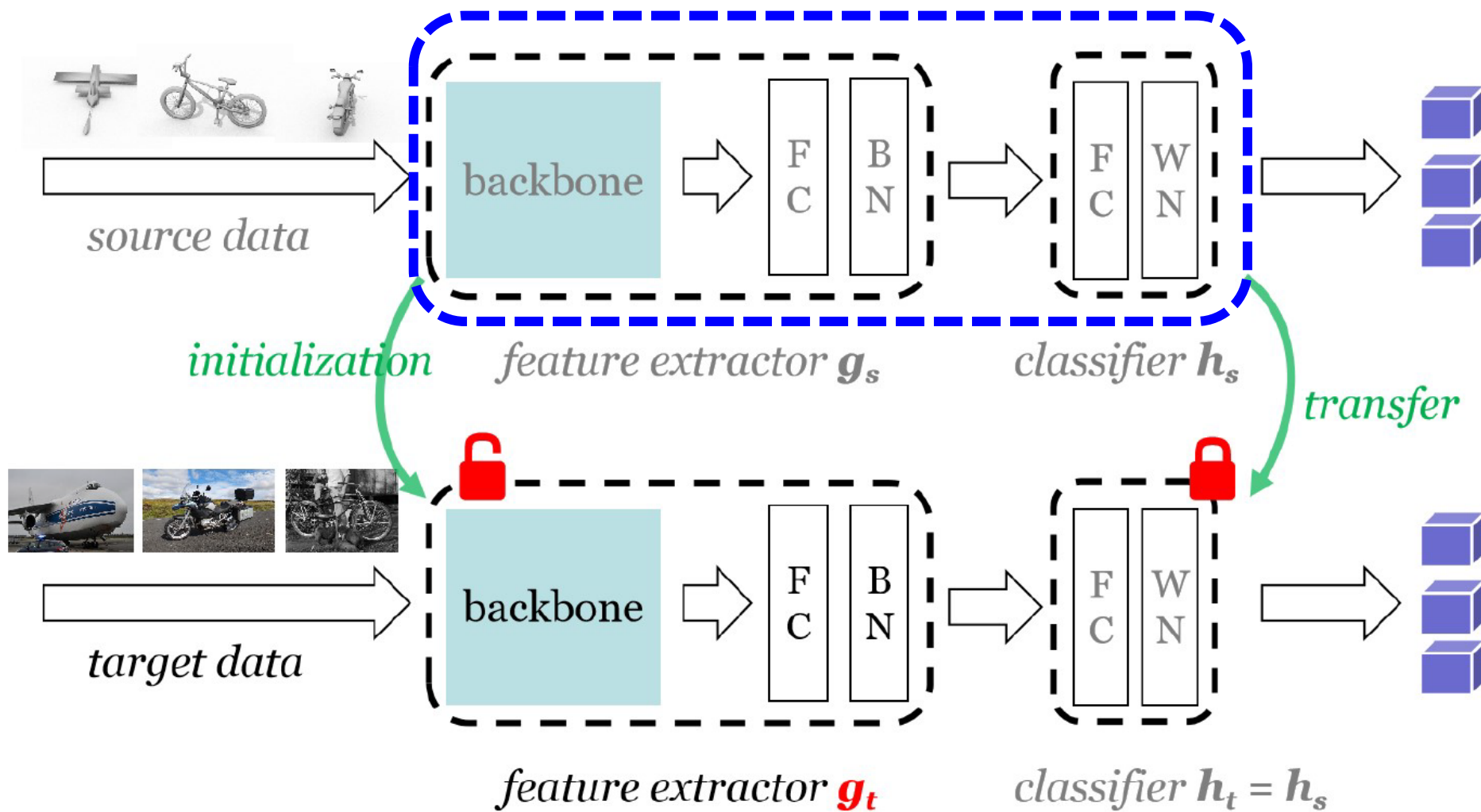
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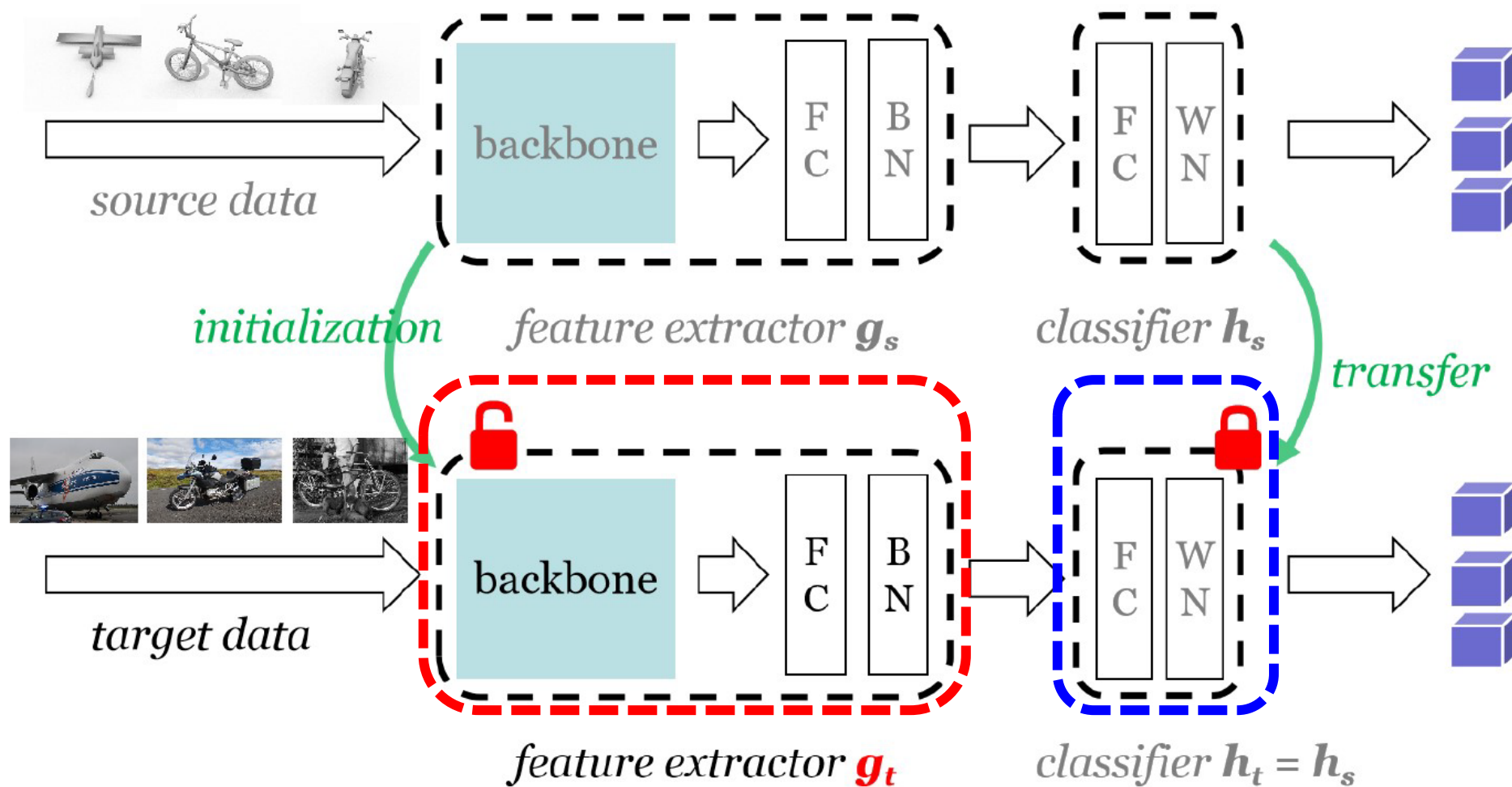
SHOT



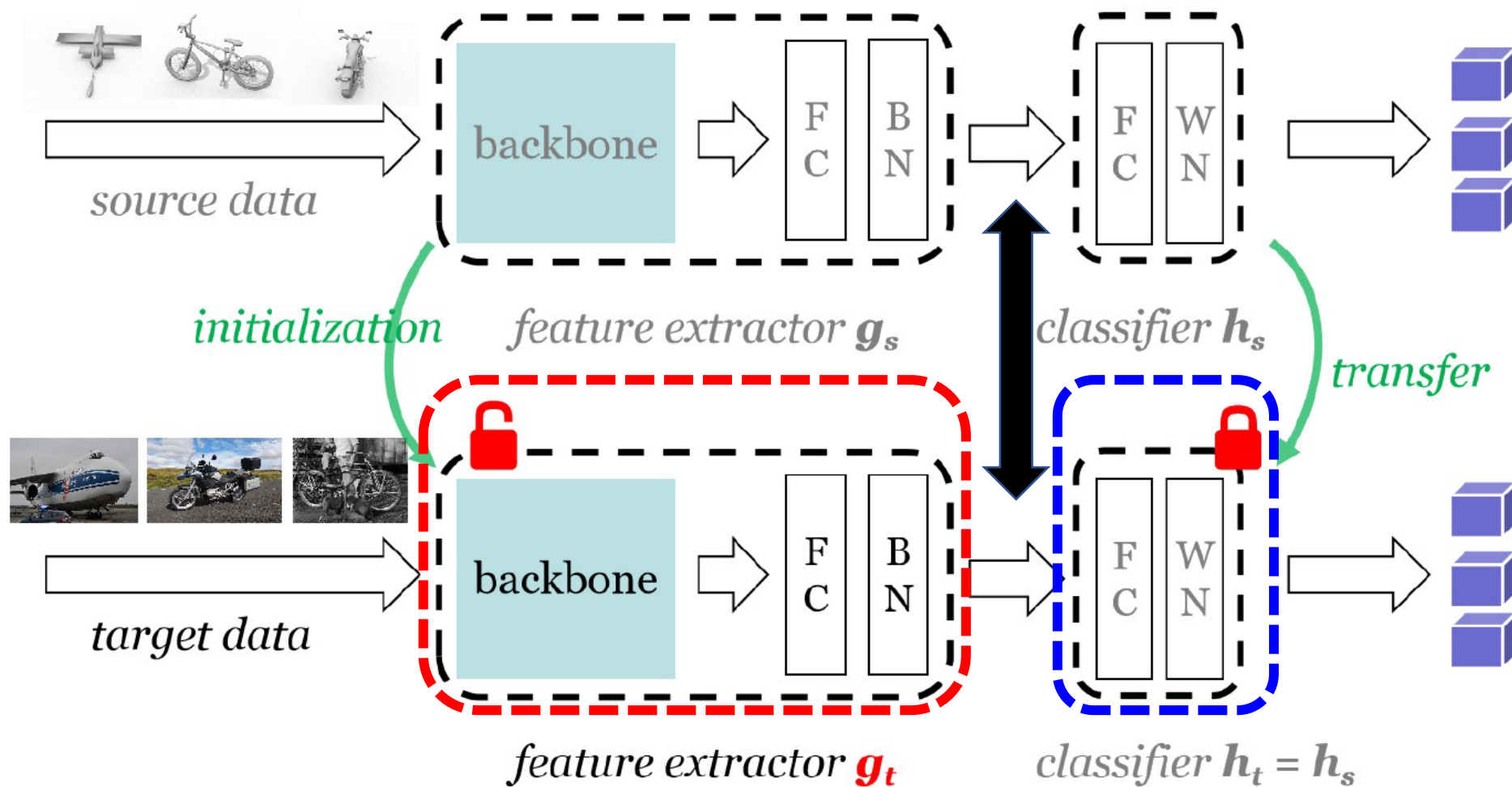
SHOT



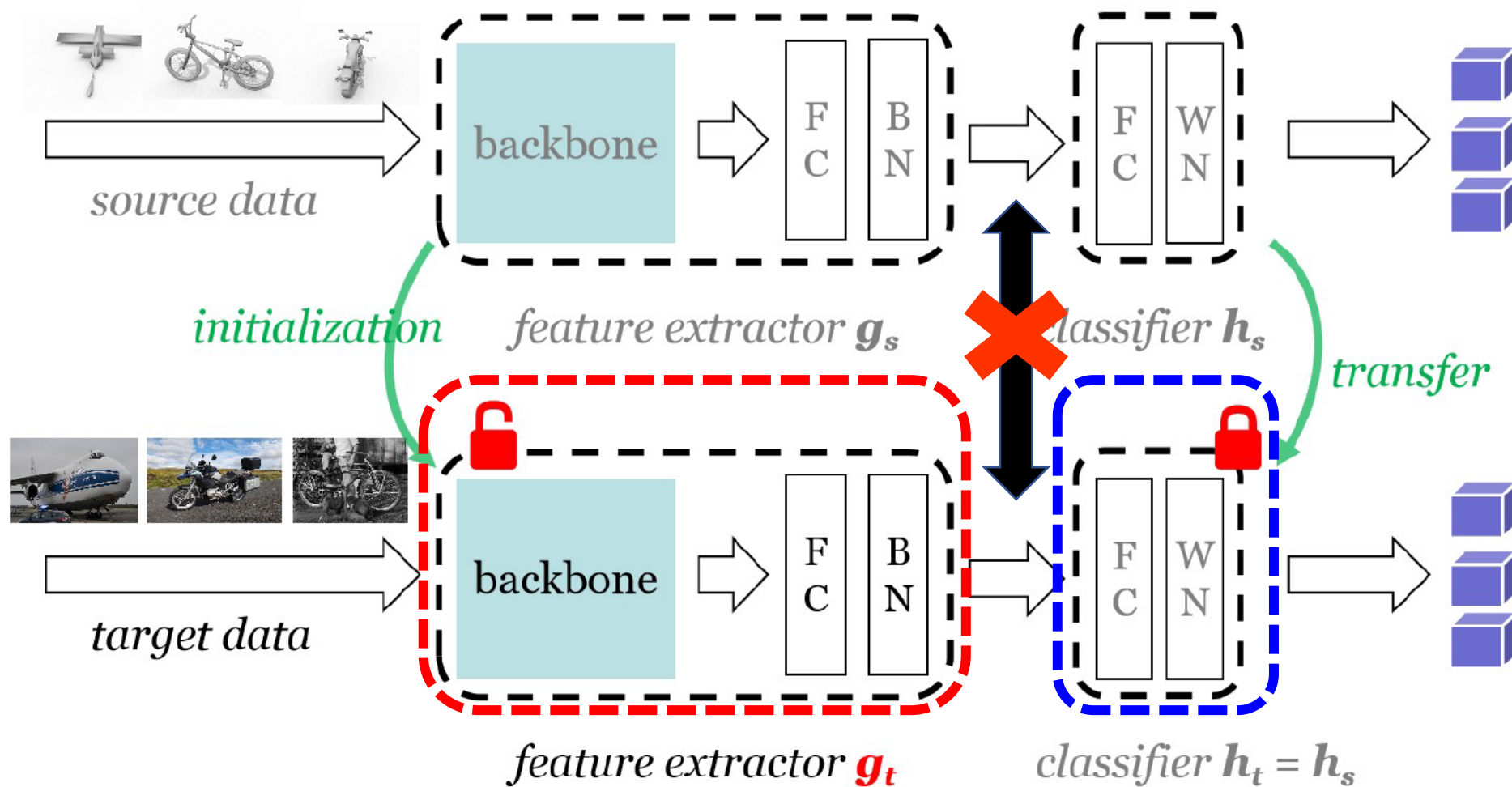
SHOT



SHOT



SHOT



SHOT

If the domain gap is mitigated,
outputs of target would be similar to
outputs of source
(one-hot encoding but differ from each other)

source data

target data

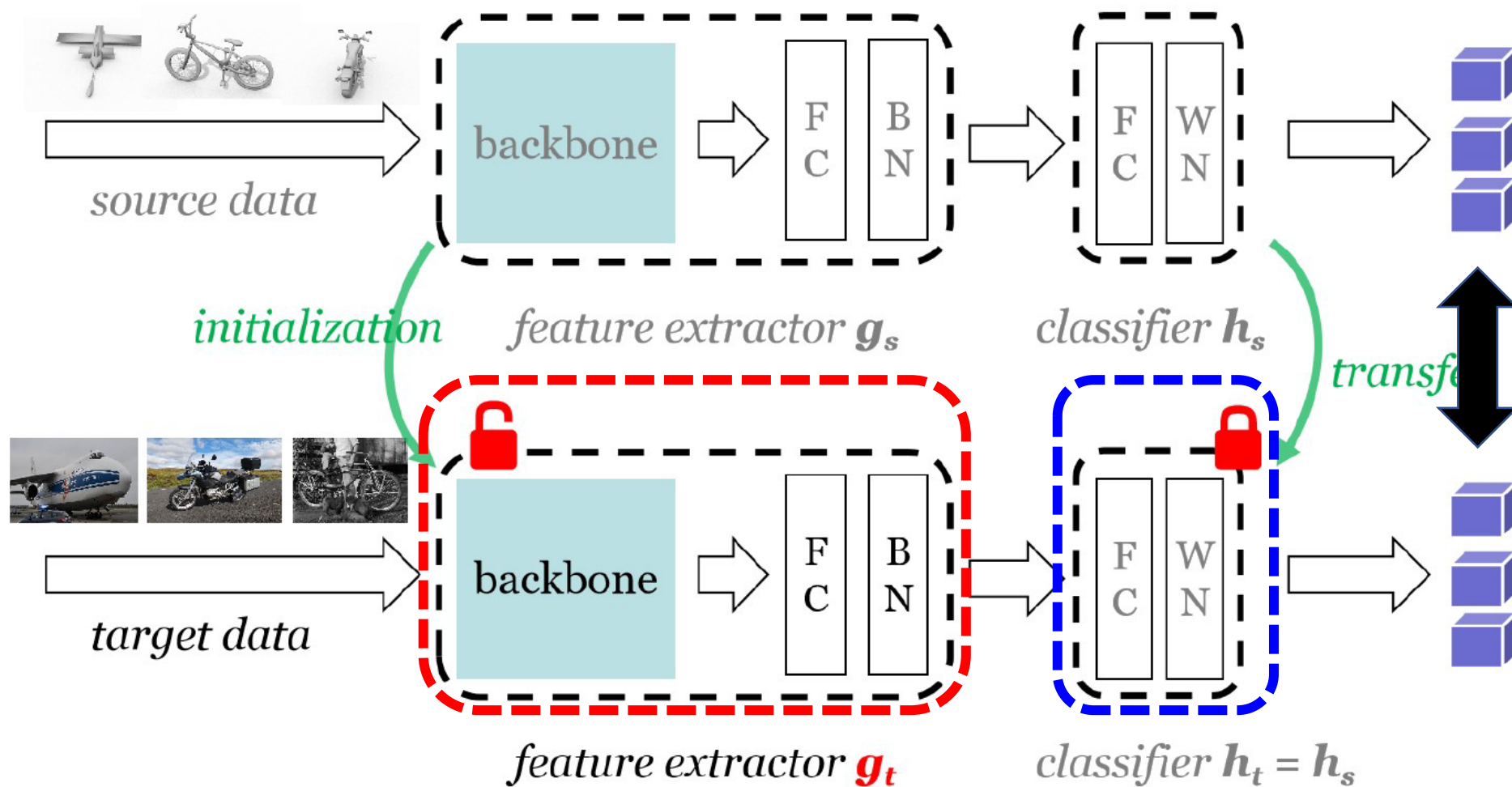
backbone

feature extractor g_t

classifier $h_t = h_s$

Transfer

SHOT



SHOT

- **Loss function**

$$\mathcal{L}_{ent}(f_t; \mathcal{X}_t) = -\mathbb{E}_{x_t \in \mathcal{X}_t} \sum_{k=1}^K \delta_k(f_t(x_t)) \log \delta_k(f_t(x_t))$$

$$\begin{aligned} \mathcal{L}_{div}(f_t; \mathcal{X}_t) &= \sum_{k=1}^K \hat{p}_k \log \hat{p}_k \\ &= D_{KL} \left(\hat{p}, \frac{1}{K} \mathbf{1}_K \right) - \log K \end{aligned}$$

$\delta_k(a)$: k th element in the soft max output of vector a

$f_t(x)$: output of data x

$\mathbf{1}_K$: K – dimensional vector with all ones

$\hat{p} : \mathbb{E}_{x_t \in \mathcal{X}_t} [\delta(f_t^{(k)}(x_t))]$

SHOT

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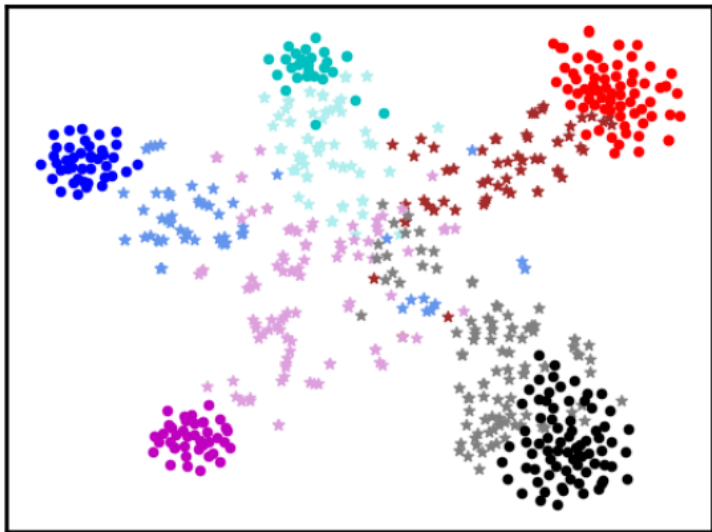
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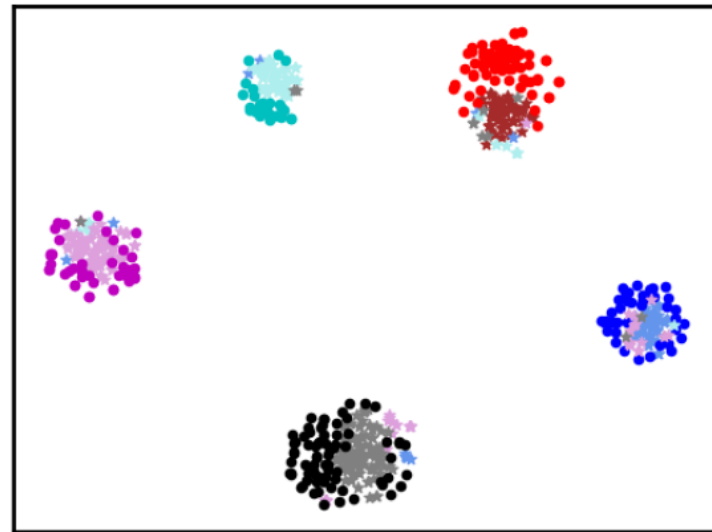
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SHOT-IM



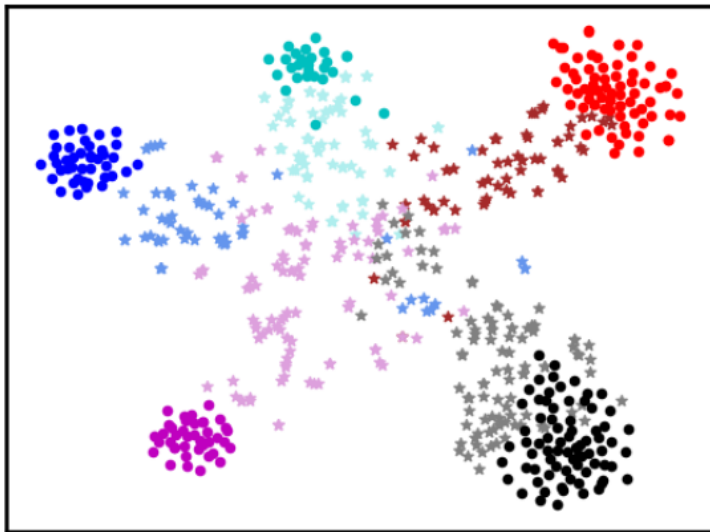
(a) Source model only



(b) SHOT-IM

Figure 1. t-SNE visualizations for a 5-way classification task. Circles in dark colors denote unseen source data and stars in light denote target data. Different colors represent different classes.

SHOT-IM



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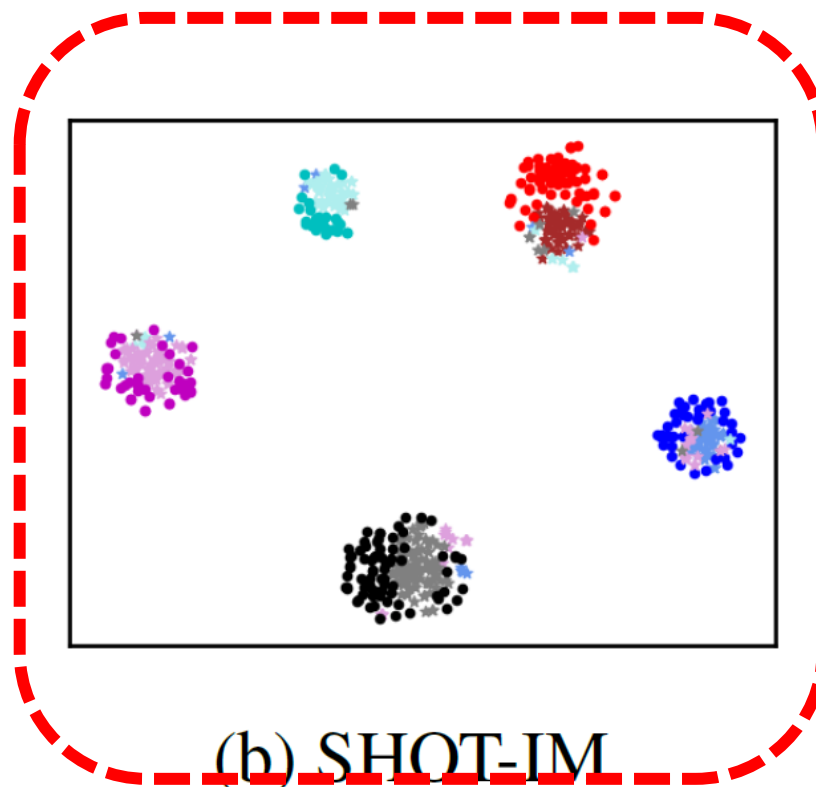
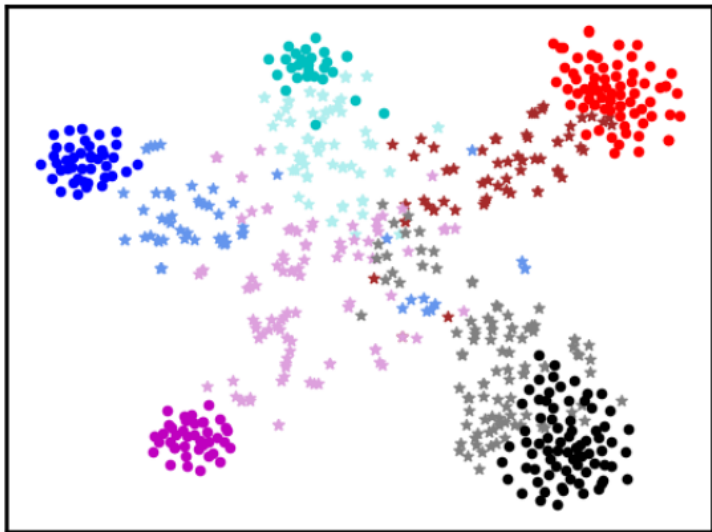


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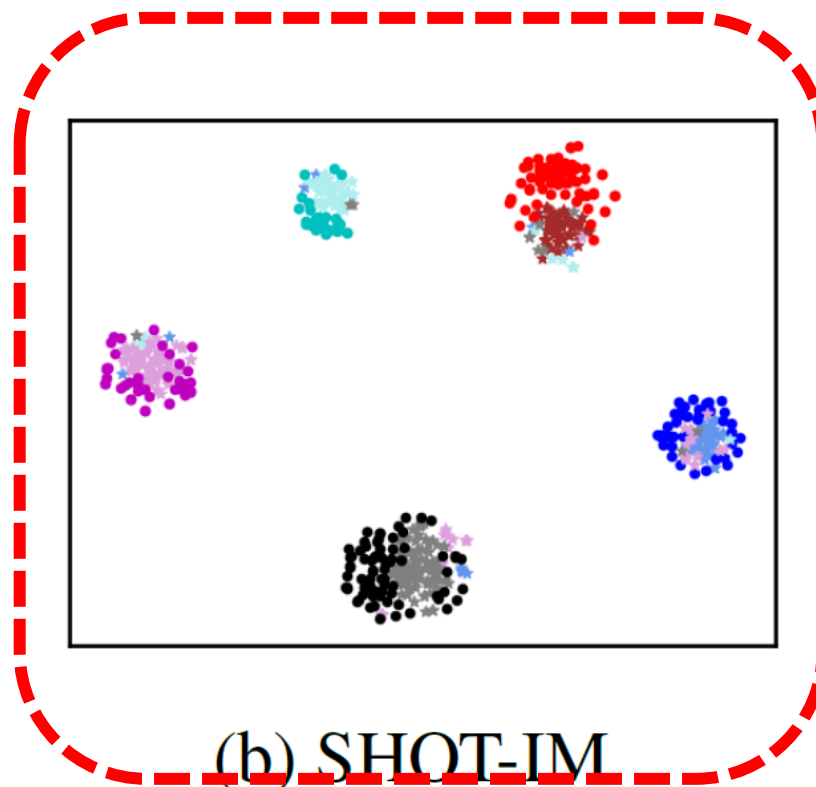


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Pseudo-Labeling

- **Self-supervised pseudo-labeling strategy**

$$c_k^{(0)} = \frac{\sum_{x_t \in \mathcal{X}_t} \delta_k(\hat{f}_t(x_t)) \hat{g}_t(x_t)}{\sum_{x_t \in \mathcal{X}_t} \delta_k(\hat{f}_t(x_t))}$$

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$D_f(a, b)$: cosine distance between a and b

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SHOT

- **Loss function**

$$\begin{aligned}\mathcal{L}(g_t) = & \mathcal{L}_{ent}(f_t; \mathcal{X}_t) + \mathcal{L}_{div}(f_t; \mathcal{X}_t) \\ & + \beta \mathbb{E}_{(x_t, \hat{y}_t) \in \mathcal{X}_t \times \hat{\mathcal{Y}}_t} \sum_{k=1}^K 1[k = \hat{y}_t] \log \delta_k(f_t(x_t))\end{aligned}$$

β : balancing hyper – parameter

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Question

4. Experiments

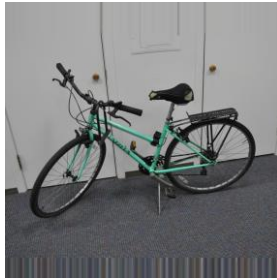
Experimental Setup

I. Datasets(Office-31)

Amazon



DSLR



Webcam



Experimental Setup

II. Methodology

- *LeNet-5(USPS/MNIST) pre-trained ResNet-50,101(Other datasets)*
 - *Label Smoothing*
 - *FC-layer replaced with bottleneck-layer(BN + Weight Normalization)*
- *Hyperparameter*
 - *64 batch size*
 - *Set β as 0.3 for all experiments (0.1 for Digits dataset).*
 - *100 epochs for training **source** / 15 epochs for training **target***

Experiments

III. Results(Office-31)

Method (Source→Target)	A→D	A→W	D→A	D→W	W→A	W→D	Avg.
ResNet-50 (He et al., 2016)	68.9	68.4	62.5	96.7	60.7	99.3	76.1
DANN (Ganin & Lempitsky, 2015)	79.7	82.0	68.2	96.9	67.4	99.1	82.2
DAN (Long et al., 2015)	78.6	80.5	63.6	97.1	62.8	99.6	80.4
CDAN+E (Long et al., 2018)	92.9	94.1	71.0	98.6	69.3	100.	87.7
rRevGrad+CAT (Deng et al., 2019)	90.8	94.4	72.2	98.0	70.2	100.	87.6
SAFN+ENT (Xu et al., 2019)	90.7	90.1	73.0	98.6	70.2	99.8	87.1
CDAN+BSP (Chen et al., 2019)	93.0	93.3	73.6	98.2	72.6	100.	88.5
CDAN+TransNorm (Wang et al., 2019)	94.0	95.7	73.4	98.7	74.2	100.	89.3
Source model only	80.8	76.9	60.3	95.3	63.6	98.7	79.3
SHOT-IM (ours)	90.6	91.2	72.5	98.3	71.4	99.9	87.3
SHOT (full, ours)	94.0	90.1	74.7	98.4	74.3	99.9	88.6

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SHOT-IM (ours)	90.6	91.2	72.5	98.3	71.4	99.9	87.3
SHOT (full, ours)	94.0	90.1	74.7	98.4	74.3	99.9	88.6

Experiments

III. Results(Office-31)

Method (Source→Target)	A→D	A→W	D→A	D→W	W→A	W→D	Avg.
ResNet-50 (He et al., 2016)	68.9	68.4	62.5	96.7	60.7	99.3	76.1
DANN (Ganin & Lempitsky, 2015)	79.7	82.0	68.2	96.9	67.4	99.1	82.2
DAN (Long et al., 2015)	78.6	80.5	63.6	97.1	62.8	99.6	80.4
CDAN+E (Long et al., 2018)	92.9	94.1	71.0	98.6	69.3	100.	87.7
rRevGrad+CAT (Deng et al., 2019)	90.8	94.4	72.2	98.0	70.2	100.	87.6
SAFN+ENT (Xu et al., 2019)	90.7	90.1	73.0	98.6	70.2	99.8	87.1
CDAN+BSP (Chen et al., 2019)	93.0	93.3	73.6	98.2	72.6	100.	88.5
CDAN+TransNorm (Wang et al., 2019)	94.0	95.7	73.4	98.7	74.2	100.	89.3
Source model only	80.8	76.9	60.3	95.3	63.6	98.7	79.3
SHOT-IM (ours)	90.6	91.2	72.5	98.3	71.4	99.9	87.3
SHOT (full, ours)	94.0	90.1	74.7	98.4	74.3	99.9	88.6

Experiments

III. Results(Office-Home)

Table 4. Classification accuracies (%) on medium-sized **Office-Home** dataset for *vanilla closed-set DA* (ResNet-50).

Method (Source→Target)	Ar→Cl	Ar→Pr	Ar→Re	Cl→Ar	Cl→Pr	Cl→Re	Pr→Ar	Pr→Cl	Pr→Re	Re→Ar	Re→Cl	Re→Pr	Avg.
ResNet-50 (He et al., 2016)	34.9	50.0	58.0	37.4	41.9	46.2	38.5	31.2	60.4	53.9	41.2	59.9	46.1
DANN (Ganin & Lempitsky, 2015)	45.6	59.3	70.1	47.0	58.5	60.9	46.1	43.7	68.5	63.2	51.8	76.8	57.6
DAN (Long et al., 2015)	43.6	57.0	67.9	45.8	56.5	60.4	44.0	43.6	67.7	63.1	51.5	74.3	56.3
CDAN+E (Long et al., 2018)	50.7	70.6	76.0	57.6	70.0	70.0	57.4	50.9	77.3	70.9	56.7	81.6	65.8
CDAN+BSP (Chen et al., 2019)	52.0	68.6	76.1	58.0	70.3	70.2	58.6	50.2	77.6	72.2	59.3	81.9	66.3
SAFN (Xu et al., 2019)	52.0	71.7	76.3	64.2	69.9	71.9	63.7	51.4	77.1	70.9	57.1	81.5	67.3
CDAN+TransNorm (Wang et al., 2019)	50.2	71.4	77.4	59.3	72.7	73.1	61.0	53.1	79.5	71.9	59.0	82.9	67.6
Source model only	44.6	67.3	74.8	52.7	62.7	64.8	53.0	40.6	73.2	65.3	45.4	78.0	60.2
SHOT-IM (ours)	55.4	76.6	80.4	66.9	74.3	75.4	65.6	54.8	80.7	73.7	58.4	83.4	70.5
SHOT (full, ours)	57.1	78.1	81.5	68.0	78.2	78.1	67.4	54.9	82.2	73.3	58.8	84.3	71.8

Experiments

III. Results(Ablation Study - Loss)

Methods / Datasets	Office	Office-Home	VisDA-C
Source model only	79.3	60.2	46.6
naive pseudo-labeling (PL) (Lee, 2013)	83.0	64.1	76.6
Self-supervised PL (ours)	87.6	68.9	80.7
$\mathcal{L}_{ent}$	83.5	55.5	63.3
$\mathcal{L}_{ent} + \mathcal{L}_{div}$	87.3	70.5	80.4
$\mathcal{L}_{ent} + \mathcal{L}_{div}$ + naive PL (Lee, 2013)	87.5	70.3	82.9
$\mathcal{L}_{ent} + \mathcal{L}_{div}$ + Self-supervised PL	88.6	71.8	82.9

Experiments

III. Results(Ablation Study - Loss)

Methods / Datasets	Office	Office-Home	VisDA-C
Source model only	79.3	60.2	46.6
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$\mathcal{L}_{ent} + \mathcal{L}_{div}$	87.3	70.5	80.4
$\mathcal{L}_{ent} + \mathcal{L}_{div} + \text{naive PL (Lee, 2013)}$	87.5	70.3	82.9
$\mathcal{L}_{ent} + \mathcal{L}_{div} + \text{Self-supervised PL}$	88.6	71.8	82.9

Experiments

III. Results(Ablation Study - Loss)

Methods / Datasets	Office	Office-Home	VisDA-C
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Experiments

III. Results(Ablation Study - Loss)

Methods / Datasets	Office	Office-Home	VisDA-C
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$\mathcal{L}_{ent} + \mathcal{L}_{div}$ + naive PL (Lee, 2013)	87.5	70.3	82.9
$\mathcal{L}_{ent} + \mathcal{L}_{div}$ + Self-supervised PL	88.6	71.8	82.9

Experiments

III. Results(Ablation Study – Architecture Component)

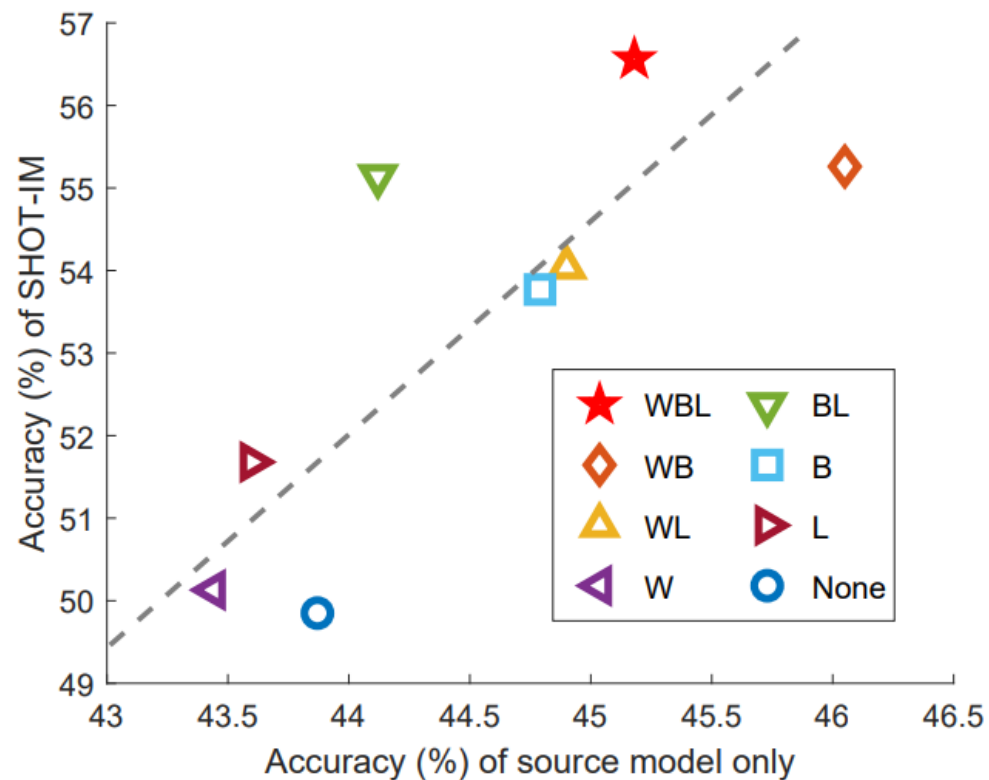


Figure 3. Accuracies (%) on the Ar→Cl task for *closed-set UDA*.
[Weight normalization/ **B**atch normalization/ **L**abel smoothing]

Experiments

III. Results(Ablation Study – Architecture Component)

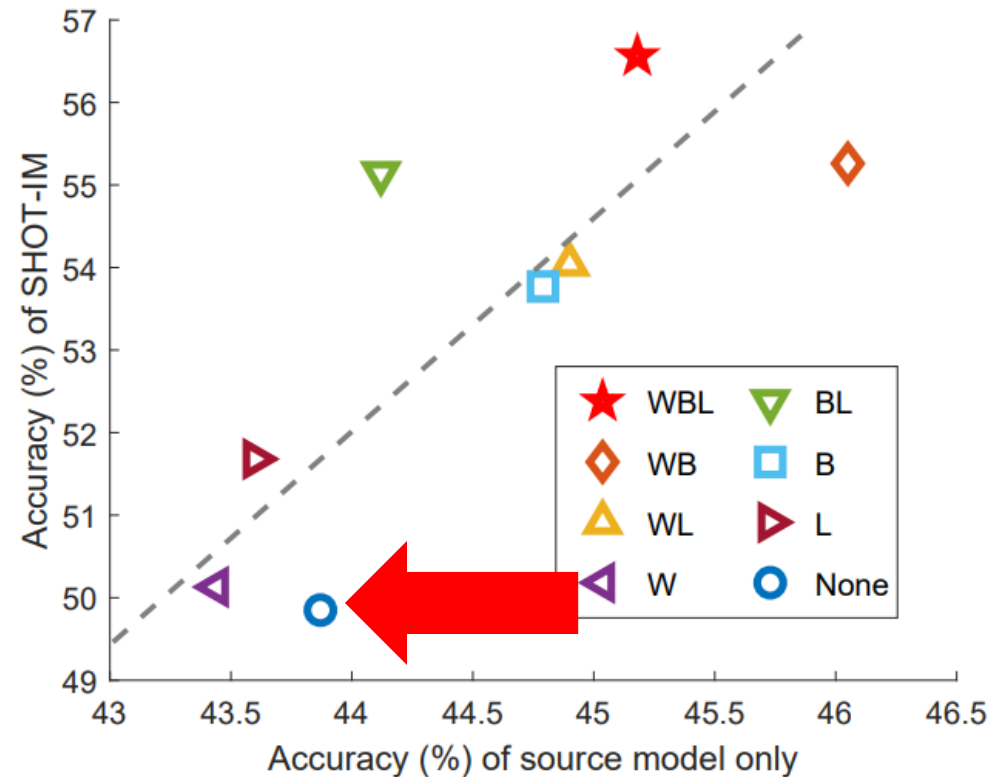


Figure 3. Accuracies (%) on the Ar→Cl task for *closed-set UDA*.
[Weight normalization/ **B**atch normalization/ **L**abel smoothing]

Question

Conclusion

- I. Practical Setting (merely needs well-trained source model)*
- II. Comparable to Source-Accessible Domain Adaptation Methods*

Thank you